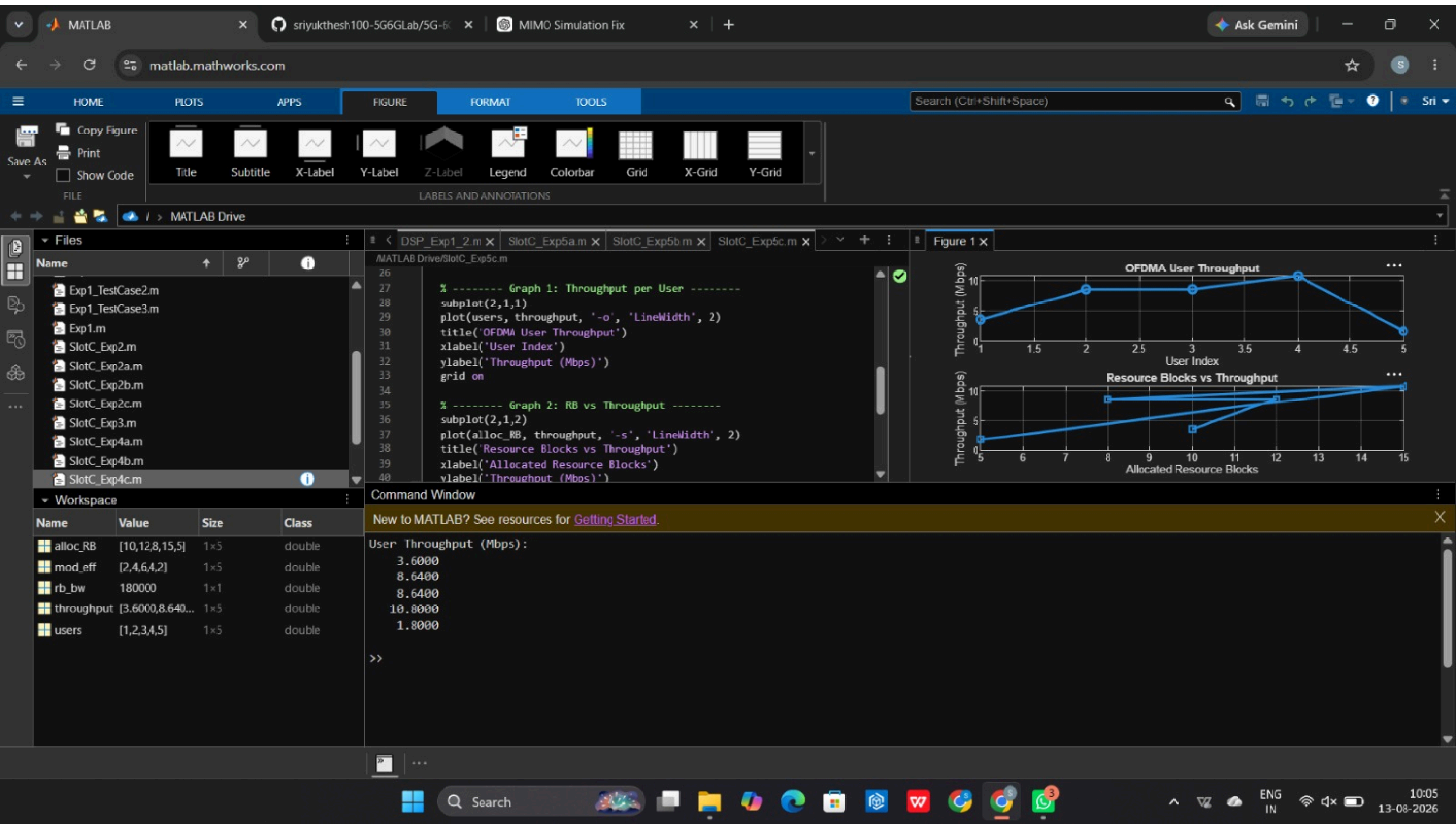
MATLAB
sriyukthesh100-5G6GLab/5G-6
MIMO Simulation Fix
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LABELS AND ANNOTATIONS
MATLAB Drive
Files
Name
Exp1_TestCase2.m
Exp1_TestCase3.m
Exp1.m
SlotC_Exp2.m
SlotC_Exp2a.m
SlotC_Exp2b.m
SlotC_Exp2c.m
SlotC_Exp3.m
SlotC_Exp4a.m
SlotC_Exp4b.m
SlotC_Exp4c.m
DSP_Exp1_2.m
SlotC_Exp5a.m
SlotC_Exp5b.m
SlotC_Exp5c.m
/MATLAB Drive/SlotC_Exp5c.m
% -------- Graph 1: Throughput per User --------
subplot(2,1,1)
plot(users, throughput, '-o', 'LineWidth', 2)
title('OFDMA User Throughput')
xlabel('User Index')
ylabel('Throughput (Mbps)')
grid on
% -------- Graph 2: RB vs Throughput --------
subplot(2,1,2)
plot(alloc_RB, throughput, '-s', 'LineWidth', 2)
title('Resource Blocks vs Throughput')
xlabel('Allocated Resource Blocks')
ylabel('Throughput (Mbps)')
Figure 1
OFDMA User Throughput
Throughput (Mbps)
User Index
Resource Blocks vs Throughput
Throughput (Mbps)
Allocated Resource Blocks
Workspace
Name Value Size Class
alloc_RB [10,12,8,15,5] 1×5 double
mod_eff [2,4,6,4,2] 1×5 double
rb_bw 180000 1×1 double
throughput [3.6000,8.640... 1×5 double
users [1,2,3,4,5] 1×5 double
Command Window
New to MATLAB? See resources for Getting Started.
User Throughput (Mbps):
3.6000
8.6400
8.6400
10.8000
1.8000
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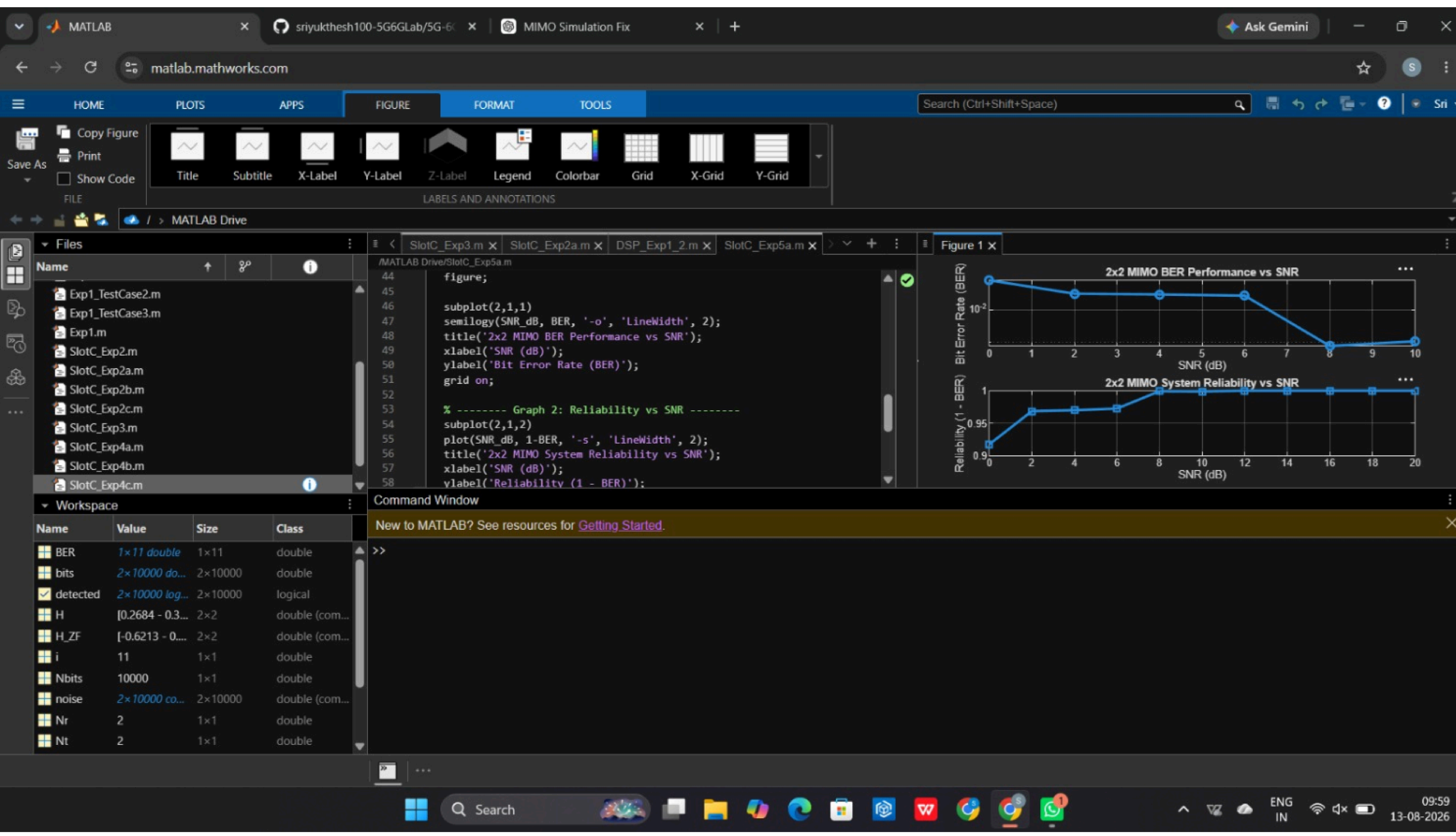
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SlotC_Exp2b.m
SlotC_Exp2c.m
SlotC_Exp3.m
SlotC_Exp4a.m
SlotC_Exp4b.m
SlotC_Exp4c.m
Workspace
Name Value Size Class
BER 1×11 double 1×11 double
bits 2×10000 do... 2×10000 double
detected 2×10000 log... 2×10000 logical
H [0.2684 - 0.3... 2×2 double (com...
H_ZF [-0.6213 - 0.... 2×2 double (com...
i 11 1×1 double
Nbits 10000 1×1 double
noise 2×10000 co... 2×10000 double (com...
Nr 2 1×1 double
Nt 2 1×1 double
SlotC_Exp3.m
SlotC_Exp2a.m
DSP_Exp1_2.m
SlotC_Exp5a.m
/MATLAB Drive/SlotC_Exp5a.m
figure;
subplot(2,1,1)
semilogy(SNR_dB, BER, '-o', 'LineWidth', 2);
title('2x2 MIMO BER Performance vs SNR');
xlabel('SNR (dB)');
ylabel('Bit Error Rate (BER)');
grid on;
% -------- Graph 2: Reliability vs SNR --------
subplot(2,1,2)
plot(SNR_dB, 1-BER, '-s', 'LineWidth', 2);
title('2x2 MIMO System Reliability vs SNR');
xlabel('SNR (dB)');
ylabel('Reliability (1 - BER)');
Figure 1
2x2 MIMO BER Performance vs SNR
Bit Error Rate (BER)
SNR (dB)
2x2 MIMO System Reliability vs SNR
Reliability (1 - BER)
SNR (dB)
Command Window
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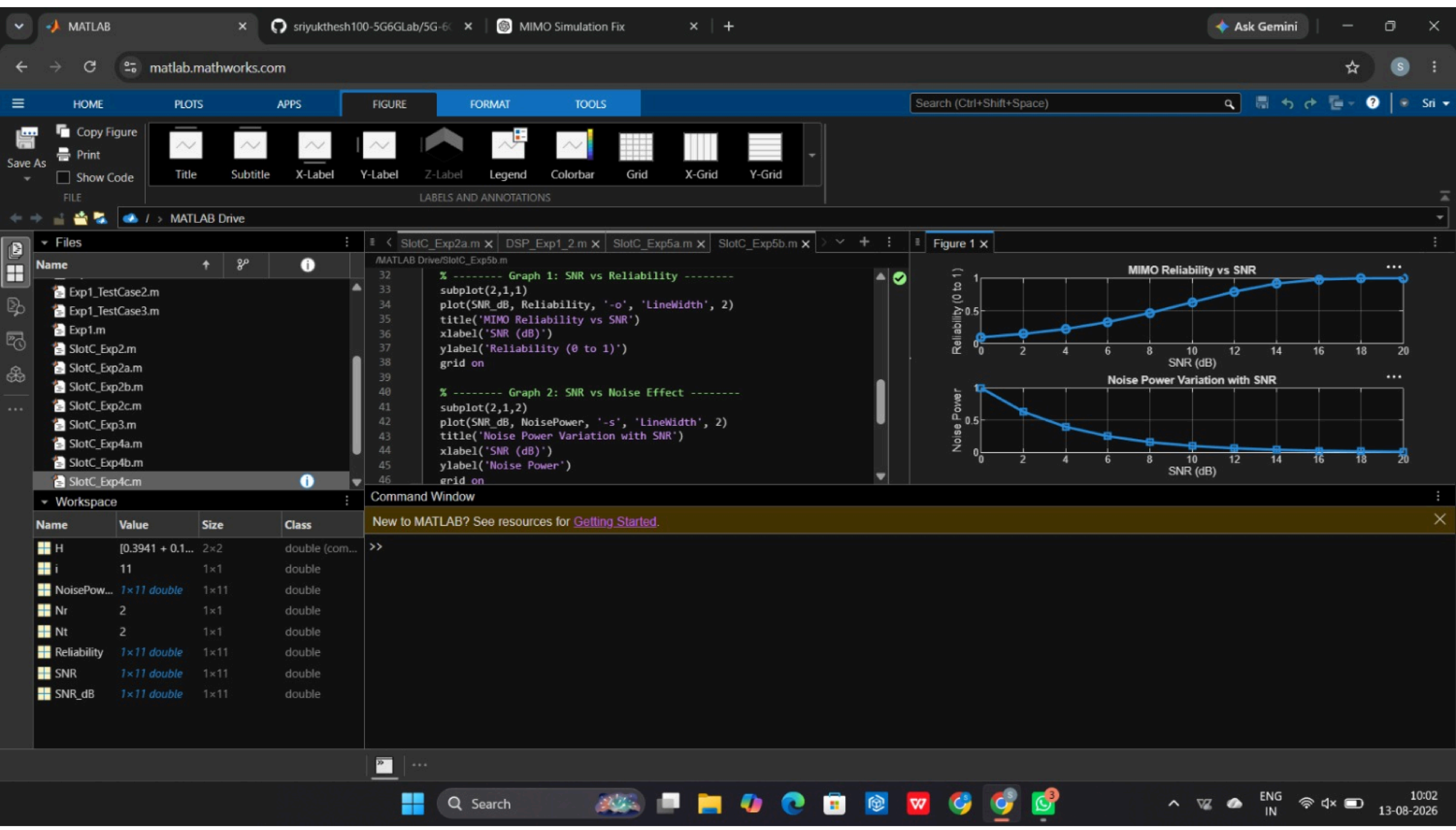

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SlotC_Exp3.m
SlotC_Exp4a.m
SlotC_Exp4b.m
SlotC_Exp4c.m
SlotC_Exp2a.m
DSP_Exp1_2.m
SlotC_Exp5a.m
SlotC_Exp5b.m
/MATLAB Drive/SlotC_Exp5b.m
% -------- Graph 1: SNR vs Reliability --------
subplot(2,1,1)
plot(SNR_dB, Reliability, '-o', 'LineWidth', 2)
title('MIMO Reliability vs SNR')
xlabel('SNR (dB)')
ylabel('Reliability (0 to 1)')
grid on
% -------- Graph 2: SNR vs Noise Effect --------
subplot(2,1,2)
plot(SNR_dB, NoisePower, '-s', 'LineWidth', 2)
title('Noise Power Variation with SNR')
xlabel('SNR (dB)')
ylabel('Noise Power')
grid on
Figure 1
MIMO Reliability vs SNR
Reliability (0 to 1)
SNR (dB)
Noise Power Variation with SNR
Noise Power
SNR (dB)
Workspace
Name Value Size Class
H [0.3941 + 0.1... 2x2 double (com...
i 11 1x1 double
NoisePow... 1x11 double 1x11 double
Nr 2 1x1 double
Nt 2 1x1 double
Reliability 1x11 double 1x11 double
SNR 1x11 double 1x11 double
SNR_dB 1x11 double 1x11 double
Command Window
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10:02
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